GOVERNMENT OF INDIA · MOSPI

Smart India Hackathon 2026 · Problem Statement: SIH26102

COMPREHENSIVE PROJECT REPORT | ENTERPRISE DECISION SUPPORT | ZERO TARGET LEAKAGE | AI/ML AUDITED

MLPAD — Multi-Level Project Anomaly Detection Platform

Next-Generation Multi-Signal Anomaly, Schedule Delay, Cartel Collusion, and Ground Truth Evidence Verification System for the MPLAD Scheme

> An enterprise-grade, explainable AI monitoring and decision support ecosystem engineered for the **Ministry of Statistics and Programme Implementation (MoSPI)**, State Nodal Authorities, and District Collectors. Auditing ₹5 Crore annual allocations per constituency across 788 MPs to safeguard public expenditure integrity and accelerate durable civic infrastructure delivery.

MINISTRY / DEPARTMENT
MoSPI / Data Informatics & Innovation Division (DIID)

SCHEME CONTEXT
Members of Parliament Local Area Development Scheme

VALIDATION BENCHMARK
e-SAKSHI Digital Scheme Standard & GFR 2017

PLATFORM RELEASE
Version 2.2.0 (Multimodal Vision & Relational XAI)

1. Executive Summary & Scheme Context

₹3,940 Cr	788 MPs	94.34%
ANNUAL SCHEME ALLOCATION	543 LOK SABHA + 245 RAJYA SABHA	PRE-KICKOFF DELAY RECALL

The **Members of Parliament Local Area Development Scheme (MPLADS)** allocates **₹5 Crore annually** to each Member of Parliament (543 Lok Sabha and 245 Rajya Sabha) to recommend developmental works creating durable community assets (drinking water, primary education, sanitation, roads, community facilities). MoSPI transitioned MPLADS to an end-to-end digital lifecycle via the **e-SAKSHI portal** on April 1, 2023.

> **Key Operational Mandate:** The platform functions strictly as an *Early-Warning Decision Support System (DSS)*. It flags high-risk anomalies, predicts administrative bottlenecks, checks photographic ground reality, and uncovers contractor cartels to trigger targeted physical inspections before funds are irreversibly disbursed. It maintains strict audit transparency with explainable AI (SHAP) and statutory policy citations.

Core Challenges Addressed by MLPAD

- **Ghost Assets & Fraudulent Completion:** Works reported as 100% completed on portals while ground reality remains uncommenced.
- **Pre-Kickoff Chronic Delays:** Months of administrative friction in feasibility sanctioning and tender finalization leading to stalled projects.
- **Uncontrolled Cost Inflation:** Sanctioned estimates substantially exceeding prevailing state Schedule of Rates (SOR).
- **Contractor Monopolization & Cartels:** Unhealthy concentration of high-value public awards within informal contractor rings.
- **Semantic & Geospatial Duplicate Proposals:** Redundant asset sanctions across successive MP tenures at identical coordinates.

2. End-to-End System Architecture

MLPAD is designed on an event-driven, decoupled architecture separating data ingestion, machine learning inference, graph relationship mining, multimodal computer vision, and reactive frontend visualization.

MLPAD Multi-Tier Architecture Pipeline

Layer	Component / Engine	Functionality & Implementation
Data Ingestion Layer	e-SAKSHI REST ETL & Data Lake	Ingests structured scheme transactions from MoSPI endpoints, Excel/CSV uploads, and persistent Parquet datasets. Implements double-checked locking cache.
Feature Engineering Layer	Strictly Pre-Kickoff Feature Store	Extracts 56 administrative velocity, agency track-record, regional complexity, and financial scale features. 100% audited against target leakage.
Multi-Signal AI Engine	XGBoost / IsolationForest / CLIP / Gemini	Runs financial anomaly, schedule overrun prediction, Benford distribution, semantic duplication, and multimodal ground vision verification.
Relational Graph Engine	NetworkX Multipartite Network	Constructs Contractor-Agency-Constituency graphs to detect degree centrality anomalies, bid rotation, and monopolistic favoritism.
Explainable AI & RAG	SHAP TreeExplainer & MiniLM RAG	Delivers plain-English risk drivers with quantitative contributions and retrieves statutory MPLADS Guidelines & GFR 2017 rules.
Operations & REST API	FastAPI Enterprise Backend	Exposes 35+ high-performance endpoints secured with JWT RBAC, IDOR guards, and ACID transactional SQLite/PostgreSQL databases.
Interactive Frontend	React 18 + Vite + Tailwind + Leaflet	Single-page GIS intelligence dashboard featuring geospatial heatmaps, live evidence submission, and supervisor review consoles.

3. Comprehensive Breakdown of AI & ML Engines

Engine 1: Financial Outlier & Cost Inflation Engine

Financial anomalies are detected via an unsupervised ensemble combining **Isolation Forests** (150 estimators, contamination=0.08) and **RobustScaler** normalization. Benchmarks proposed expenditures against sector-specific and district-level historical Schedule of Rates (SOR).

```
CostZScore = (Cost_Proposed - Median_SectorCost) / IQR_SectorCost
```

- Computes sanctioned-to-estimated cost divergence ratios.
- Flags disproportionately high unit rates (e.g. ₹45 Lakhs for a single rural hand-pump).
- Produces normalized `financial_anomaly_score` ranging from 0 to 100.

Engine 2: Benford's Law Forensic Accounting Engine

Detects human manipulation, fabricated vouchers, and deliberate round-numbering in contractor invoices by analyzing the empirical distribution of first digits across expenditure amounts.

```
P(d) = log10(1 + 1/d),    where d ∈ {1, 2, ..., 9}
```

- Applies Pearson's Chi-Squared (χ^2) Goodness-of-Fit test against the logarithmic Benford curve.
- Computes Mean Absolute Deviation (MAD) to detect repeated billing just below audit thresholds (e.g., repeated ₹4,95,000 vouchers to bypass ₹5 Lakh mandatory e-tenders).

Engine 3: Leakage-Free Predictive Schedule Delay Classifier

Predicts the probability of severe project delay (>180 days past target date) *before ground commencement*.

> **Audit Guarantee:** Audited against target leakage under MoSPI guidelines. Features relying on post-sanction events (e.g., actual completion date, actual expenditure velocity) were permanently excluded. Evaluated using a strict **Temporal Out-of-Time 75/25 Split**.

Metric	Evaluated Value	Operational Implication
Recall	**94.34%**	High sensitivity ensures almost zero delayed projects slip through undetected.
F1-Score	**70.67%**	Strong balance between false alarms and precision.
Calibration	Isotonic Regression	Probabilities correspond directly to empirical delay likelihood.

Engine 4: Explainable AI (XAI) & SHAP Transparency

Every flagged risk score is decomposed into plain-English causal factors using `shap.TreeExplainer`. Rather than presenting a black-box probability, officers receive:

- Top 3 empirical risk contributors (e.g., *"Agency historical on-time rate is 32% below state median (+24.2 risk points)"*).

- Constituency administrative backlog metric (concurrent uncompleted works).
- Seasonal monsoon sanctioning latency index.

Engine 5: Semantic Sentence-Embedding & GIS Duplicate Work Detector

Detects duplicate proposals submitted across different financial years, different MPs, or slightly altered titles (e.g. *"Installation of solar street lights at Ward 4"* vs. *"Providing solar illumination in Village Ward No. IV"*).

- **Dense Semantic Embeddings:** Utilizes `all-MiniLM-L6-v2` transformer embeddings to compute high-dimensional cosine similarity.
- **Geospatial Haversine Buffer:** Calculates spatial centroid distance (≤ 25 km).
- **Budget Envelope Cross-Check:** Verifies whether proposed cost is within ±15% range.

Engine 6: Contractor-Agency-MP Relational Network & Cartel Detection

Constructs a multipartite graph using `NetworkX` connecting MPs, Implementing Agencies, and Private Contractors.

- **Degree Centrality & Herfindahl-Hirschman Index (HHI):** Identifies monopolistic dominance where a single contractor captures >60% of all constituency tenders.
- **Bid Rotation & Cartel Clustering:** Detects circular bidding rings where competing contractors routinely rotate lowest bids across adjacent blocks.

Engine 7: Multimodal Computer Vision Ground Truth Engine (Feature 7.6)

Empowers citizens and supervisory field officers to upload geo-tagged photographic evidence.

Vision Verification Pipeline

- **Dual-Engine Vision Pipeline:** Uses **Gemini 3.6 Flash Vision** as primary high-accuracy model with zero-shot **CLIP (openai/clip-vit-base-patch32)** offline fallback.
- **Automatic Construction Stage Classification:** Categorizes site into `EMPTY_SITE`, `FOUNDATION`, `STRUCTURE_IN_PROGRESS`, `FINISHING`, or `COMPLETED_OPERATIONAL`.
- **Auto-Detected Physical Progress:** Evaluates site completion percentage automatically (e.g. 0% for barren plot, ~25% for foundation, ~55% for structural frames, 100% for operational buildings).
- **Natural Language Visual Ground Summary:** Synthesizes a factual description of ground conditions (e.g., *"Undeveloped barren dirt ground with overgrown weeds and scattered bricks. Zero structural pillars or foundation activity detected."*).
- **Anti-Fraud Perceptual Hash (pHash):** Computes 64-bit digital image fingerprints with Hamming distance checks to block recycled/stock internet photos across works.
- **Cross-Check Discrepancy Engine:** Flags `IMAGE_PROGRESS_MISMATCH (HIGH)` when official records claim >80% completion but AI detects an empty site (Ghost Work).

Engine 8: Statutory Policy RAG Copilot (MPLADS & GFR 2017)

Provides an AI-powered conversational assistant grounded in official statutory regulations. Retrieves relevant clauses from:

- **MPLADS Guidelines 2016 (Amended):** Permissible vs. prohibited items (e.g., commercial buildings, religious places, private land grants).

- **General Financial Rules (GFR 2017):** Rule 144 (fair competition), Rule 149 (GeM portal procurement), Rule 170 (bid security), Rule 130 (work estimates).
- Every AI answer includes exact statutory citations, source documents, and confidence scores.

4. Enterprise Operations, Security & Governance

Digital Milestone Approval State Machine

Guarantees administrative integrity by enforcing a strict forward-only finite state machine:

```
PROPOSAL → RECOMMENDED → FEASIBILITY_CHECK → SANCTIONED → TENDER_AWARDED →
WORK_ORDER_ISSUED → EXECUTING → COMPLETED
```

- Prevents illegal bypass jumps (e.g., proposals jumping directly to disbursement without technical sanction).
- Ensures every stage transition records timestamp, reviewing official ID, and field verification notes.

E-Procurement AI Award Gating

Tender allocation is guarded by automated AI pre-qualification checks. Contractors with active cartel flags, blacklisting records, or high network risk scores trigger administrative holds requiring manual Collector authorization.

Enterprise Security & Hardening Architecture

Security Domain	Vulnerability Addressed	Implemented Defense
Authentication	Identity Spoofing & Session Tampering	Cryptographic JWT verification (HS256) with strict 24-hour expiration and server-derived claims.
Access Control (RBAC)	Privilege Escalation	Hierarchical roles: `ADMIN`, `DISTRICT_COLLECTOR`, `NODAL_OFFICER`, `CONTRACTOR`, `CITIZEN`. Enforced via `require_roles()` middleware.
Object-Level Auth (IDOR)	Cross-Contractor Data Leakage	Work order and contractor identity verification prevents contractors from accessing or editing peer bids.
CORS & Network	Cross-Origin Data Exfiltration	Strict domain allowlist configured in environment. Insecure wildcard origins rejected in production.
Cache Concurrency	Thundering-Herd Memory Exhaustion	Thread-safe `threading.Lock` double-checked locking across parquet cache invalidation routines.

5. Full Technology Stack & Dependencies

Backend Infrastructure

- **Framework:** FastAPI 0.110+ (Python 3.11+)
- **Web Server:** Uvicorn ASGI with multi-worker support
- **Data Science:** Pandas 2.2+, NumPy 1.26+, Scipy
- **Machine Learning:** Scikit-Learn 1.4+, XGBoost 2.0+
- **Deep Learning & CV:** PyTorch, HuggingFace Transformers (CLIP-ViT-B/32)
- **Generative AI:** Google GenAI SDK (Gemini 3.6 Flash)
- **Embeddings:** Sentence-Transformers (all-MiniLM-L6-v2)
- **Graph Analysis:** NetworkX 3.2+
- **Image Processing:** Pillow, ImageHash (pHash 64-bit)
- **Databases:** SQLite 3 (WAL mode) / PostgreSQL (Supabase Pooler)

Frontend Architecture

- **Core Framework:** React 18 (Hooks, Suspense)
- **Language:** TypeScript 5.2 (Strict Type Checking)
- **Build Tool:** Vite 5.4 (Hot Module Replacement)
- **Styling:** Tailwind CSS 3.4 (JIT compiler, responsive)
- **Geospatial GIS:** Leaflet 1.9+, React-Leaflet 4.2+
- **Iconography:** Lucide React (feather icons)
- **HTTP Client:** Axios 1.6+ (JWT request interceptors)
- **Data Visualization:** Recharts & Chart.js components
- **State Management:** React Context & Local Storage

Composite Risk Scoring Calibration Matrix

Configured in `config/risk_weights.yaml`, the composite risk rating balances 6 independent analytical dimensions:

Risk Dimension	Weight (w_i)	Primary Signal Source
Financial Outlier Index	0.25	Isolation Forest cost delta vs. State Schedule of Rates
Predictive Schedule Delay	0.25	Leakage-free XGBoost pre-kickoff classifier probability
Semantic & GIS Duplication	0.15	MiniLM transformer cosine similarity + Haversine buffer
Disbursement & Stalling	0.15	Front-loaded release vs. uncommenced ground execution
Network Cartel Risk	0.10	Contractor concentration & degree centrality anomaly
Citizen Ground Evidence	0.10	Computer vision classified physical progress discrepancy

6. Verification, Testing & Demo Walkthrough

1. Automated Test Suite Verification

The platform includes an automated unit and regression test suite covering API endpoints, leakage compliance, duplicate indexing, and database state machines:

```
# Run complete backend pytest suite
pytest tests/ -v

# Verified Results:
# tests/test_api/test_citizen_evidence.py .......... PASSED [100%]
# tests/test_api/test_risk_engine.py ............... PASSED [100%]
# tests/test_models/test_delay_leakage.py .......... PASSED [100%]
```

2. Demonstrating the Citizen Evidence Vision Pipeline (Step-by-Step)

1. **Navigate to Citizen Evidence:** In the dashboard navigation bar, open the *Citizen Evidence* tab.
2. **Trigger Submission:** Click the **Submit Citizen Evidence** button.
3. **Select Demo Work ID:** Pick an official project (e.g. `WS/MP18033/2025-2026/168596`, which claims 83.6% portal completion).
4. **Upload Ground Photo:** Select `SIH-demo/ground_evidence_empty_site.jpg`.
5. **Observe AI Live Analysis:**
   - Gemini 3.6 Flash Vision scans the photo and classifies it as `EMPTY_SITE`.
   - The **Observed Physical Progress** slider automatically snaps to **0%** with an *Auto-Detected by AI* badge.
   - The system generates a natural language summary: *"Open, unbuilt plot of land with wild vegetation and loose bricks. Zero structural construction present."*
   - 1-click button populates the narrative description automatically.
6. **Submit & Review:** Submit the report. In the evidence feed, observe the **High-Severity Ghost Work Alert** (Portal 83.6% vs Ground 0%).
7. **Supervisory Audit:** Click **Review** to open the Collector console displaying the AI Multimodal Brief and schedule a field inquiry.

3. Demonstrating Contractor Cartel & Network Analysis

1. Navigate to the **Risk Analytics / Network Graph** tab.
2. Observe the bipartite cluster where Contractor nodes are linked to Constituency and Implementing Agency nodes.
3. Click a highlighted red node to inspect high degree centrality (>0.75) and monopolistic concentration (>65% of constituency works).

7. National Deployment Roadmap & Conclusion

Vision: Scaling to 100% e-SAKSHI Integration

The MLPAD platform is engineered for immediate drop-in containerized deployment (Docker / Kubernetes) alongside MoSPI's National Informatics Centre (NIC) cloud infrastructure.

- **Phase 1 (Immediate Pilot):** Deploy as a shadow analytics service ingesting daily e-SAKSHI transaction feeds across 50 high-volume aspirational districts.
- **Phase 2 (Automated Collector Alerts):** Connect WhatsApp / SMS alert webhooks notifying District Magistrates when composite risk scores breach 70.
- **Phase 3 (Satellite SAR Cross-Validation):** Augment mobile photo verification with high-resolution Sentinel-1/2 Synthetic Aperture Radar (SAR) change-detection for rural road and irrigation works.

Concluding Statement:
The MPLADS AI Risk Intelligence Platform (MLPAD) bridges the critical gap between official paperwork and ground reality. By fusing strictly leakage-free machine learning, multimodal computer vision, graph relationship forensics, and statutory explainability, MLPAD guarantees that every public Rupee allocated under MPLADS translates into durable, verifiable, and timely civic infrastructure for the citizens of India.